

Learner Guide

World Happiness — what to do, why, and where you are in the process | ★★★★★☆

This guide shows one sound way to complete the scenario, the reasoning behind each step, and which stage of the data analysis process each step belongs to. Assessors look for evidence that you followed the process in order. The trickiest parts here are reshaping the population file and reconciling country names — get those right and the analysis is straightforward. Work the task yourself first, then check your method here. Screenshots are illustrative mock-ups; Excel/Power Query is shown.

The data analysis process — follow it in order:

Identify → Collect → Clean/Prepare → Analyse → Visualise → Interpret/Report

Each step is tagged with its stage. Most of the marks on this scenario sit in Clean/Prepare — the unpivot and the name reconciliation.

Scenario 1 — Data Gathering and Validation

Step 1 — Identify what you need

STAGE 1 · IDENTIFY

What you do: Confirm the requirement: three sources (two happiness years plus the population file), the goal of one combined set with a Year column and population joined on country name. Note the known issues — the population file is transposed with title rows, and country names differ between sources.

Why start here? Spotting that the population file needs reshaping, and that names won't all match, before you begin means you plan the work rather than hitting surprises mid-task.

Step 2 — Bring each source in as a connection

STAGE 2 · COLLECT

What you do: Import all three sources as **connection only**.

1. Data → Get Data → From File → the CSV or Excel file.
2. At the load prompt, choose **only create connection**.

Import Data

×

Select how you want to view this data in your workbook.

☐ Table
 ☐ PivotTable Report
 ☐ PivotChart
 ☒ **Only Create Connection**

↑ Keeps each source as a query to be cleaned and reshaped.

OK

Cancel

Figure 1 — Choose **Only Create Connection** so each source stays as a query.

Why connection only? The population file needs real reshaping before it's usable; a connection-only query lets you do that in Power Query without dumping the messy layout onto a sheet.

Step 3 — Reshape the population file (unpivot)

STAGE 3 · CLEAN / PREPARE

What you do: Double-click the population query to open it. Remove the title rows, promote the real header, then **unpivot** the country columns so each row becomes one country-year-population.

1. Remove Top Rows to strip the World Bank title rows.
2. Use First Row as Headers to promote the country names.
3. Select the country columns and choose Unpivot Columns.
4. Rename the new columns to **Country** and **Population**; keep only the year(s) you need.

Power Query Editor — Population

Unpivot Columns

(select the country columns, then Unpivot to turn them into rows)

Before — countries across columns, years down rows:

Year	Finland	United Kingdom	Japan	... (260 more)
2017	5,508,...	66,058,...	126,8...	
2018	5,515,...	66,460,...	126,5...	

↓ Unpivot ↓

After — one row per country-year (tidy format):

Year	Country	Population
2018	Finland	5,515,525
2018	United Kingdom	66,460,344

Tidy long format is what merges and charts need — one value per row.

Figure 2 — Unpivoting turns the wide, transposed layout into a tidy one row per country-year.

Why unpivot? The data arrives "wide" — countries spread across hundreds of columns — which can't be joined or charted. Unpivoting reshapes it into "long" (tidy) format, one value per row, which is the shape every merge and chart expects. This reshape is the single biggest skill this scenario tests.

Step 4 — Append the happiness years and add Year

STAGE 3 · CLEAN / PREPARE

What you do: Add a **Year** column to each happiness query (2018 / 2019), check the column names match across both, then **Append Queries** to stack them into one table. Remove duplicates and check for missing values.

Why a Year column, and why append? The two files have identical columns for different years, so they stack (append). Without a Year column the rows are indistinguishable once combined — and Year is exactly what Scenario 2's year-over-year comparison depends on.

Step 5 — Join population and reconcile country names

STAGE 3 · CLEAN / PREPARE

What you do: Merge the population data onto the combined happiness set on **country name**. Then check for unmatched rows — any country with a **null** population after the join has a name mismatch to fix.

Merge result — country names that did not match (Population = null)

Happiness country	Population country (WDI)	Population
Swaziland	Eswatini	null ⚠
Congo (Kinshasa)	Congo, Dem. Rep.	null ⚠
North Macedonia	Macedonia, FYR	null ⚠
United Kingdom	United Kingdom	66,460,344 ✓
Finland	Finland	5,515,525 ✓

⚠ A null population after the join flags a name mismatch — these must be reconciled (e.g. Swaziland = Eswatini).

Figure 3 — A null population after the join flags a name mismatch (e.g. Swaziland = Eswatini).

Why check for nulls after a join? A text-key join only matches exact spellings, so differently-named countries silently drop out. Filtering for null population is the quickest way to surface them. Reconcile the handful that differ (Swaziland → Eswatini; Congo (Kinshasa) → Congo, Dem. Rep.; North Macedonia → Macedonia) so no country is lost. Save as **Happiness_Population_Combined.xlsx**.

Scenario 2 — Data Analysis

Step 6 — Year-over-year and benchmark comparisons

STAGE 4 · ANALYSE

What you do: Calculate each country's score change 2018 → 2019, then the global average and the UK against it.

Expected results: global average happiness rose slightly, from **5.37** (2018) to **5.41** (2019). The UK scored **7.19** then **7.05** — above the global average in both years, but down year-on-year.

Step 7 — G7 comparison

STAGE 4 · ANALYSE

What you do: Average the seven G7 countries' scores and compare to the global average (AVERAGEIF on a G7 flag, or a filtered PivotTable).

Expected result: the G7 average for 2019 is about **6.70** — well above the global average of 5.41, as you'd expect for large advanced economies.

Step 8 — Happiness vs population (scatter)

STAGE 5 · VISUALISE

What you do: Plot happiness score against population on a scatter, using a **log scale** for population.

Why a log scale? Populations range from under a million to over a billion — on a linear axis almost every country bunches against the left edge. A log scale spreads them out so any pattern is visible. Expect little relationship: population and happiness are essentially uncorrelated.

Step 9 — Correlation with other variables

STAGE 4 · ANALYSE

What you do: Use `=CORREL( . . . )` (or a correlation matrix) between happiness and each factor, and a scatter for the strongest.

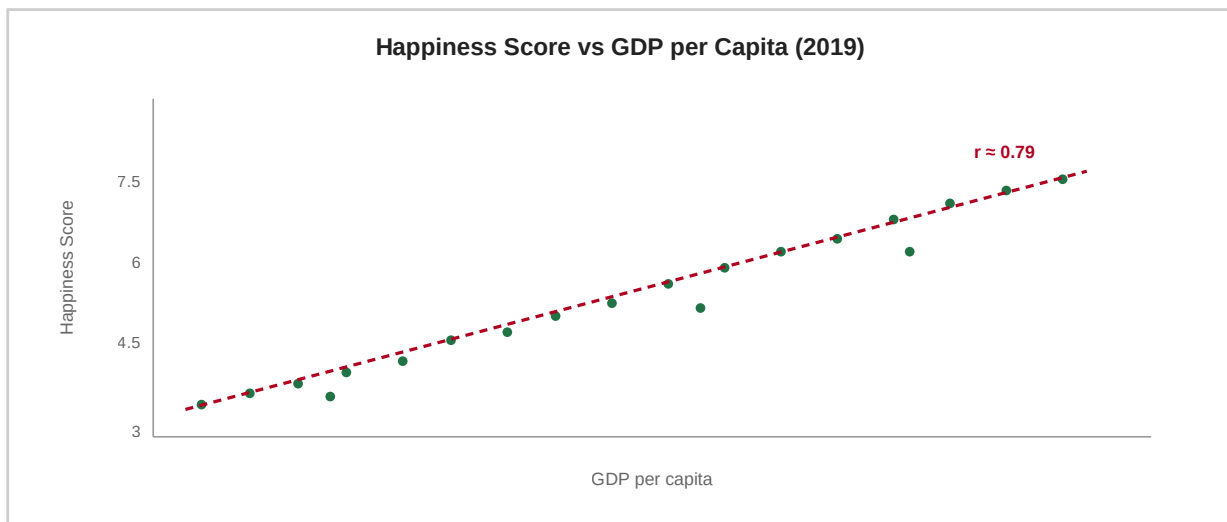


Figure 4 — Happiness vs GDP per capita: a clear positive relationship.

Expected results (2019 correlations with happiness score):

Variable	Correlation (r)
GDP per capita	0.79 (strongest)
Healthy life expectancy	0.78
Social support	0.78
Freedom to make life choices	0.57
Perceptions of corruption	0.39
Generosity	0.07 (weakest)

Why these matter: GDP, healthy life expectancy and social support are the strongest correlates of happiness; generosity barely relates to it. Remember correlation is association, not cause — wealthier countries score higher, but that doesn't prove wealth alone causes happiness.

Step 10 — Interpret and report

STAGE 6 · INTERPRET / REPORT

What you do: Summarise the findings for senior management beneath your charts, then save the workbook with its tables and charts intact.

"Global average happiness edged up from 5.37 to 5.41 between 2018 and 2019. The G7 (avg 6.70) and the UK (7.05) sit well above the world average, though the UK dipped year-on-year. Happiness correlates most strongly with GDP per capita ($r \approx 0.79$), healthy life expectancy and social support (both ≈ 0.78), and barely at all with generosity (≈ 0.07) or population. This suggests prosperity, health and strong social networks are the factors most associated with national happiness."

Marker's checklist — process followed in order

- **Identify:** three sources, goal and known issues (transposed file, name mismatches) confirmed first.
- **Collect:** data brought in as **connection only** (not opened); double-clicked to open.
- **Clean/Prepare:** population unpivoted to tidy format; happiness years appended with a Year column; population joined; name mismatches reconciled (no null populations left).
- **Analyse:** year-over-year change; UK vs global; G7 vs global; correlations across variables.
- **Visualise:** log-scale population scatter; happiness-vs-GDP scatter.
- **Interpret/Report:** findings summarised, correlation-not-causation noted; saved as .xlsx.